Sheth L.U.J and Sir M.V. College
Python Project

Google colab notebook link:
https://colab.research.google.com/drive/1TmPRr1bZPGfXjCjHUdtYWL32m9ogTV2o#scrollTo=LnsubdWzIBj5

```
import datetime #library to display date and time
from IPython.display import clear_output  #from google colab clears the previous output
import time  #module for time
import os

# --- COLORS --- (ANSI color codes)
GREEN = "\033[92m"
BLUE = "\033[94m"
PURPLE = "\033[35m"
RED = "\033[91m"
RESET = "\033[0m"

# ---DATABASE ---
if not os.path.exists("readme.txt"):
        with open("readme.txt", "w") as f: # w = WRITE MODE - creates file
            f.write("Welcome to PYNUX! built by Tasmiya Jahagirdar")
USERNAME = "Admin"
PASSWORD = "2627@admin"

# --- LOGIN SYSTEM ---
clear_output()
print(f"{BLUE}==================================={RESET}")
print(f"{GREEN} PYNUX - Secure Edition built by Tasmiya Jahagirdar {RESET}")
print(f"{BLUE}==================================={RESET}\n")

user = input("Username: ") .strip()
pwd = input("Password:") .strip()

if user!= USERNAME or pwd!= PASSWORD:
    print(f"\n{RED}Access Denied! Wrong password.{RESET}")
else:
    print(f"\n{GREEN}Login Successful! Welcome {user}{RESET}")
    time.sleep(1) # Pauses the execution of the programm for 1 second
    clear_output()
```

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```
    print(f"{BLUE}==================================={RESET}")
    print(f"{GREEN} Welcome to PYNUX!  {RESET}")
    print(f"{PURPLE} Type 'help'  to see commands{RESET}")
    print(f"{BLUE}==================================={RESET}\n")

    while True:
        command = input(f"{GREEN} PYNUX> {RESET}").lower() .strip()

        if command == "help" :
            print(f"""
{PURPLE}--- AVAILABLE COMMANDS ---{RESET}
{BLUE}ls{RESET} - list all files
{BLUE}notepad{RESET} - create a new file
{BLUE}read{RESET} - read a file
{BLUE}delete{RESET} - delete a file
{BLUE}calc{RESET} - calculator
{BLUE}time{RESET} - show time & date
{BLUE}clear{RESET} - clear screen
{BLUE}shutdown{RESET} - exit OS
            """)

        elif command == "ls":
            print(f"\n{BLUE}--- Your Files ---{RESET}")
            if not files:
                print("No files found.")
            else:
                for f in files:
                    print(f"{GREEN}- {f}{RESET}") #Lists all the files

        elif command == "notepad":
            name = input("File name: ")
            content = input("Content: ")
            with open(name, "w") as f: # REAL FILE HANDLING: w = write,
creates real file on disk
                f.write(content)
            print(f"{GREEN}File '{name}' saved!{RESET}")

        elif command == "read":
            name = input("Enter the name of the file to read: ")
            if os.path.exists(name): # REAL FILE HANDLING: check if
file exists
```

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```
                with open(name, "r") as f: # REAL FILE HANDLING: r =
read mode
                    data = f.read()
                print(f"\n{PURPLE}--- {name}
---{RESET}\n{files[name]}{PURPLE}--------------{RESET}")
            else:
                print(f"{RED}File not found!{RESET}")

        elif command == "delete":
            name = input("File name to delete: ")
            if os.path.exists(name):
                os.remove(name) # FILE HANDLING: os.remove()
                print(f"{RED}File '{name}' deleted!{RESET}")
            else:
                print(f"{RED}File not found!{RESET}")

        elif command == "calc":
            try:   # a risky command that have chances of error
                exp = input("Enter calc (e.g. 5*5): ")
                print(f"{GREEN}Result: {eval(exp)}{RESET}")#eval()
prints an expression like a srting
            except:  #If there is error in try the programm does not
crashes insead runs except
                print(f"{RED}Invalid calculation!{RESET}")

        elif command == "time":
            print(f"{PURPLE}{datetime.datetime.now()}{RESET}")

        elif command == "clear":
            clear_output()
            print(f"{GREEN}Screen Cleared - PYNUX{RESET}\n")

        elif command == "shutdown" :
            print(f"\n{RED}Shutting down PYNUX!!... Bye
{user}!{RESET}")
            break


        else:
            print(f"{RED}Unknown command. Type 'help'{RESET}")
```

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